

Technical Architecture Document

Media Asset Management (MAM) Platform Built on Nuxeo

2.1 Executive Summary

This document defines the technical architecture for the Media Asset Management (MAM) platform built on the Nuxeo Platform. It covers the platform stack, deployment options, storage architecture, infrastructure requirements, integration points, high availability, and security considerations.

The architecture is designed to be scalable, resilient, and cost-effective — supporting everything from a single-node deployment for small newsrooms to a multi-node high-availability cluster for large broadcasters.

2.2 Platform Stack

2.2.1 Core Components

Component	Technology	Version	Purpose
Core Platform	Nuxeo Platform	2025 LTS	Media Asset Management engine
Runtime	Java (JDK)	JDK 21 (Azul Zulu or Oracle)	Application runtime
Application Server	Apache Tomcat	10.1	Servlet container (bundled with Nuxeo)

Component	Technology	Version	Purpose
Framework	Jakarta EE	10	Enterprise Java specifications
Database	PostgreSQL	16	Primary document storage
Search Engine	Elasticsearch	Latest supported	Full-text indexing and search
Cache & Messaging	Redis	Latest supported	Distributed caching and message queuing
Binary Storage	Filesystem or S3-compatible	—	Media file storage

Compatibility Note: Nuxeo 2025 LTS officially supports JDK 21 (Azul's Zulu OpenJDK or Oracle's JDK) and PostgreSQL 16.

2.2.2 Architecture Overview

The Nuxeo Platform follows a modern, modular architecture built on Java with an OSGi-like component model. Key architectural layers include:

Layer	Description	Technologies
Presentation Layer	Web UI and REST APIs	Nuxeo Web UI, JavaScript, REST/GraphQL
Application Layer	Business logic and workflows	Java, OSGi bundles, Nuxeo Studio
Persistence Layer	Document storage and indexing	PostgreSQL, Elasticsearch, Redis
Storage Layer	Binary file storage	Filesystem, S3, Glacier

2.2.3 Database Options

The Nuxeo Repository supports multiple database backends:

Database	Support Level	Use Case
PostgreSQL	Recommended ▾	Production deployments
MySQL	Supported ▾	Alternative production
Oracle	Supported ▾	Enterprise environments
MongoDB	Supported ▾	NoSQL alternative

The default configuration for Nuxeo Repository uses a SQL Database and a filesystem storage, as well as an Elasticsearch index.

2.3 Deployment Architecture

2.3.1 Deployment Options

Option	Best For	Complexity	Reference
Docker	Development and simple production	Low	Recommended approach
Kubernetes + Helm	Scalable, HA production deployments	Medium-High	Official Helm Chart available
ZIP Distribution	Traditional on-premise installations	Medium	Legacy/enterprise environments

2.3.2 Docker Deployment

Quick Start:docker run --name nuxeo -p 8080:8080 \

```
-v /path/to/nuxeo/data:/var/lib/nuxeo \
```

```
-v /path/to/nuxeo/log:/var/log/nuxeo \
```

```
docker.packages.nuxeo.com/nuxeo/nuxeo:2025
```

Why Docker is Recommended:

- Simplifies installation and dependency management
- Enables consistent environments across development and production
- Supports easy scaling and orchestration

2.3.3 Kubernetes Deployment with Helm

The Nuxeo Helm Chart allows installation of the Nuxeo Platform on a Kubernetes cluster.

Add the Helm Repository: `helm repo add nuxeo`

`https://packages.nuxeo.com/repository/helm-releases-public/`

Install the Chart: `helm install my-release nuxeo/nuxeo`

Deployment Modes:

- **Single Node:** Simple, single-instance deployment
- **API-Worker Architecture:** Separates API and worker nodes for scalability

2.3.4 Production Environment Components

A production-ready Nuxeo cluster includes:

Component	Purpose	Minimum for HA
Load Balancers	Traffic distribution and failover	2
Reverse Proxies	Security and request routing	2 (Apache or Nginx)

Component	Purpose	Minimum for HA
Nuxeo Server Nodes	Application processing	2+
Database Cluster	Document storage	1+ (with replication)
Elasticsearch Cluster	Search and indexing	3 nodes
Redis Cluster	Caching and messaging	3 nodes
File System / S3	Binary storage	Shared storage

2.4 Storage Architecture

2.4.1 Binary Storage (Blob Storage)

By default, Nuxeo stores all blobs on the filesystem in a lock-free, cluster-aware manner, based on the "Content-addressable storage" paradigm.

Alternative Storage Options:

- **Amazon S3:** Store blobs in S3 using the [S3BinaryManager](#)
- **Other Cloud Storage:** Azure, Google Cloud Storage via plugins

Configuration Example (Multiple Blob Providers):

```
org.nuxeo.ecm.core.storage.sql.S3BinaryManager
YOUR_AWS_ACCESS_KEY_ID
YOUR_AWS_SECRET_ACCESS_KEY
my-nuxeo-bucket
us-east-1
```

> **Important:** Always define different paths when using local blob providers. When using Amazon S3 (or any other cloud provider), always define different `bucket_prefix` (or container prefix) if using the same bucket.

2.4.2 Storage Tiers

Tier	Storage Type	Use Case	Cost
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Hot	Filesystem or S3 Standard	Frequently accessed content	High
Warm	S3 Standard-IA / One Zone-IA	Less frequently accessed content	Medium
Cold	S3 Glacier	Long-term archival	Low

Supported S3 Storage Classes:

Nuxeo supports S3 Standard-IA (`STANDARDIA`), One Zone-IA (`ONEZONEIA`), and Glacier Instant Retrieval (`GLACIER_IR`) storage classes.

2.4.3 Cold Storage (Archival)

The **Nuxeo Cold Storage addon** allows optimization of storage costs by storing archived content in long-term storage leveraging AWS's Glacier service.

Requirements:

- * Standard S3 bucket for the Nuxeo instance
- * `s3:RestoreObject` permission on the bucket to restore documents from Glacier

Features:

- * Content remains searchable and previewable even in cold storage

- * Archived content can be restored on demand
- * Automatic archival policies based on age or metadata

2.4.4 Document Storage (Database)

Nuxeo stores documents and their property values either in a database (VCS) or in a NoSQL database (DBS).

Default Configuration:

- * PostgreSQL for document metadata
- * Embedded Elasticsearch instance running in the same JVM as Nuxeo (development)
- * Separate Elasticsearch cluster for production

2.5 Infrastructure Requirements

2.5.1 Minimum Requirements (Development/Quick Start)

Resource	Requirement
CPU	2 cores
RAM	2 GB
Disk Space	1 GB (minimum)

2.5.2 Production Requirements (Small Deployment)

Component	CPU	RAM	Storage

	Nuxeo Server		4 vCPU		8-16 GB		50 GB	
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	PostgreSQL		2 vCPU		4-8 GB		100 GB+	
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	Elasticsearch		2 vCPU		4-8 GB		100 GB+	
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	Redis		1 vCPU		2-4 GB		10 GB	
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2.5.3 Production Requirements (Large Deployment)

	Component		CPU		RAM		Storage	
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	Nuxeo Server (2+ nodes)		8+ vCPU each		16-32 GB each		100 GB+ each	
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	PostgreSQL Cluster		8+ vCPU		16-32 GB		1 TB+	
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	Elasticsearch Cluster (3+ nodes)		8+ vCPU each		16-32 GB each		500 GB+ each	
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	Redis Cluster (3+ nodes)		2 vCPU each		4-8 GB each		20 GB+ each	
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	Binary Storage		—		—		10 TB+ (S3/Glacier)	
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2.5.4 Software Requirements

	Software		Version		Notes	
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	JDK		JDK 21 (Azul Zulu or Oracle)		Required for development and production	
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	PostgreSQL		16		Recommended database	
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	Elasticsearch		Latest supported		Search engine	
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	Redis		Latest supported		Cache and messaging	
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| **Docker** | Latest | For containerized deployment |

| **Kubernetes** | 1.24+ | For orchestrated deployment |

2.5.5 Operating System Support

| OS | Support Level |

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| **Linux** | **Recommended** (production) |

| **Windows** | Supported |

| **macOS** | Supported (development only) |

2.6 Integration Points

2.6.1 REST API

The Nuxeo REST API provides HTTP/HTTPS access to the full Nuxeo Platform.

Capabilities:

- * CRUD operations on documents, assets, users, groups, directories

- * Workflow and task management

- * Search via NXQL and page providers

- * OAuth2 resource management

Best For:

- * Remotely integrating portals, workflow engines, ESBs, and custom applications

- * Building custom frontend applications

* Third-party system integration

2.6.2 Client SDKs

Nuxeo provides client SDKs for multiple programming languages:

SDK	Language	Use Case
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Java Client	Java	Server-side integration
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JavaScript Client	JavaScript	Web applications
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Python Client	Python	Scripting and automation
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NET Client	C#	.NET applications
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PHP Automation Client	PHP	PHP applications
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iOS/Android	Swift/Kotlin	Mobile applications
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2.6.3 YouTube Publishing Integration

****Method:**** Nuxeo Media Publishing addon

****Setup Steps:****

1. Install the Nuxeo Media Publishing addon
2. Configure OAuth2 with Google Developers Console
3. Set up YouTube as a service provider in Admin Center
4. Test single channel publishing

****Features:****

- * Direct video publishing from the Nuxeo UI
- * Support for multiple YouTube channels
- * Metadata mapping (title, description, tags, category)

2.6.4 Transcoding Integration

****Method:**** Nuxeo Conversion Service + FFmpeg

****Capabilities:****

- * Automatic proxy generation on upload
- * Transcoding to multiple formats (MP4, WEBM, etc.)
- * Thumbnail/storyboard extraction
- * Custom transcoding profiles

****Dependencies:****

- * FFmpeg installed on the server
- * `ccextractor` for closed caption extraction (optional)

2.6.5 Storage Integration

****Method:**** Blob Manager Configuration

****Supported Storage:****

- * Local filesystem (default)
- * Amazon S3 (via `S3BinaryManager`)
- * Google Cloud Storage

- * Microsoft Azure Blob Storage

- * Any S3-compatible storage (MinIO, Ceph)

2.7 High Availability & Scalability

2.7.1 High Availability Architecture

The standard Nuxeo cluster architecture providing high availability is composed of:

Component	Minimum for HA	Purpose
Load Balancers	2	Traffic distribution with sticky sessions
Reverse Proxies	2 (Apache/Nginx)	Security and request routing
Nuxeo Server Nodes	2+	Application processing
Elasticsearch Cluster	3 nodes	Search and indexing
Redis Cluster	3 nodes	Caching and messaging

2.7.2 Scalability

Scaling Out Processing:

- * Easy to scale out processing — simply add more Nuxeo Server nodes

- * Each new node increases processing capacity

Scaling Storage:

- * Partition data between "live/fresh" and "archive" for cost optimization

- * Use cold storage for older, less frequently accessed content

****Reference Architecture for Fault Tolerance:****

- * Kafka cluster (at least 2 nodes) for stream processing
- * Zookeeper cluster for coordination
- * Eliminates single points of failure (SPOF)

2.8 Security Architecture

2.8.1 Authentication & Authorization

Feature	Implementation
Authentication	Username/password, LDAP, SSO, OAuth2
Authorization	Role-based access control (RBAC)
API Security	OAuth2 token-based authentication

2.8.2 Data Security

Layer	Security Measure
In Transit	TLS 1.2+ encryption
At Rest (Database)	Database encryption (configurable)
At Rest (Storage)	S3 server-side encryption
Audit Logging	All user actions logged

2.8.3 Network Security

* Reverse proxies protect each Nuxeo server node

* Load balancers handle SSL termination

* Three-layer DMZ architecture supported

2.9 Monitoring & Operations

2.9.1 Monitoring Tools

Tool	Purpose
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Prometheus	Metrics collection
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Grafana	Dashboards and visualization
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ELK Stack	Log aggregation and analysis
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Nuxeo Audit	User action logging
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2.9.2 Key Metrics to Monitor

Metric	Purpose	Alert Threshold
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CPU Usage	Server health	> 80%
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Memory Usage (JVM Heap)	Application health	> 85%
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Disk Space	Storage capacity	> 80%
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Database Connections	Database health	> 80% of pool
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Elasticsearch Health	Search availability	Red status
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	Redis Health		Cache availability		Connection failures	
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2.9.3 Backup & Disaster Recovery

	Component		Backup Frequency		Retention	
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	Database (PostgreSQL)		Daily		30 days	
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	Elasticsearch Indexes		Weekly		30 days	
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	Binary Storage (S3)		Versioning enabled		Configurable	
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	Configuration Files		On change		Indefinite	
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2.10 Performance Considerations

2.10.1 JVM Tuning

	Parameter		Recommended Value		Notes	
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	Heap Size		2-4 GB (minimum)		Adjust based on workload	
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	Garbage Collector		G1GC		Default for JDK 21	
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	Metaspace		256 MB		Default is sufficient	
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2.10.2 Database Optimization

- * Use PostgreSQL 16 for best performance

- * Enable connection pooling

- * Index frequently queried fields

* Regular `VACUUM` and `ANALYZE`

2.10.3 Elasticsearch Optimization

* Dedicated Elasticsearch cluster for production

* At least 3 nodes for high availability

* Configure appropriate heap size (50% of available RAM, max 32GB)

2.10.4 Binary Storage Optimization

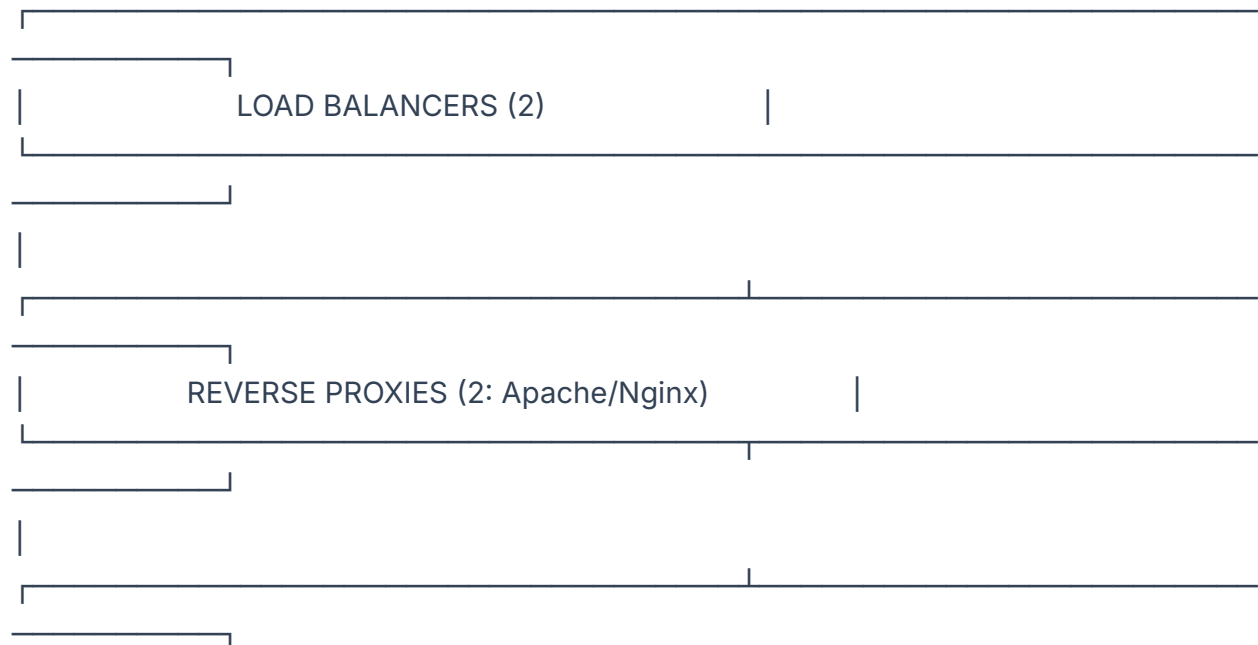
* Use S3 for scalable, cost-effective storage

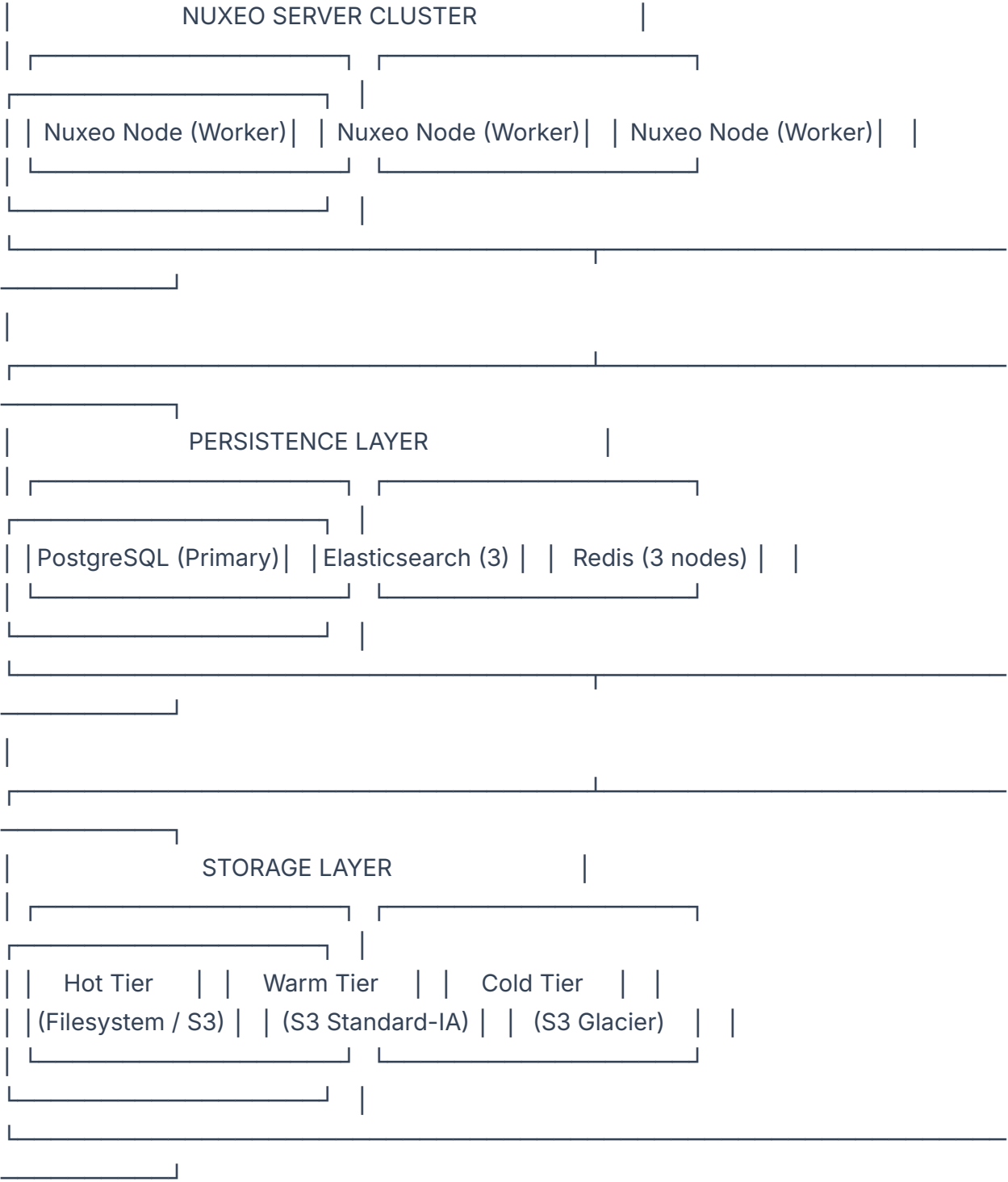
* Implement storage tiering (hot/warm/cold)

* Use content-addressable storage for deduplication

2.11 Architecture Diagrams

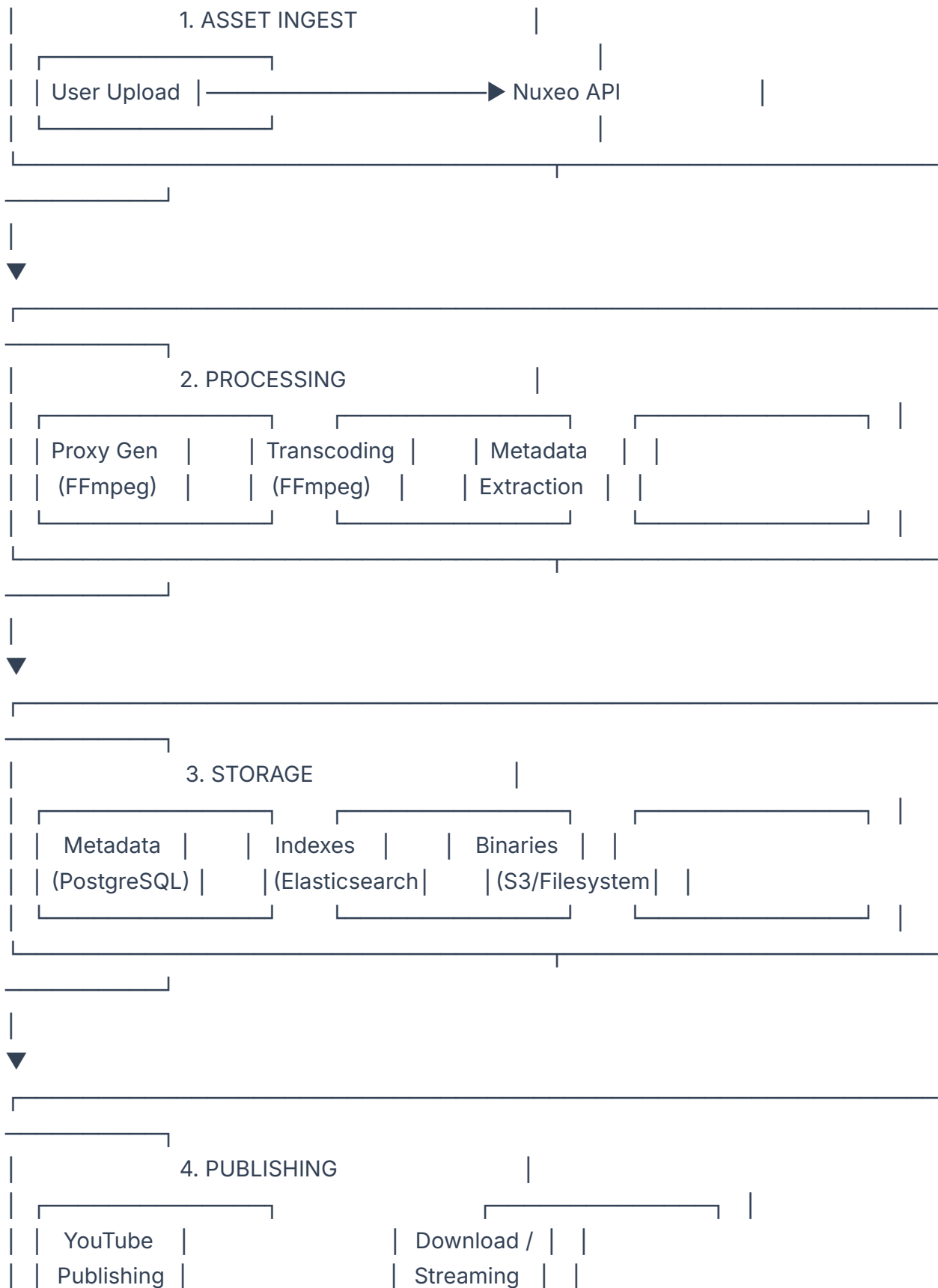
2.11.1 High-Level Architecture





2.11.2 Data Flow





2.12 Technology Selection Rationale

Component	Selection	Rationale
Nuxeo Platform	Open-source, Apache 2.0	Allows proprietary forking; proven in enterprise
JDK 21	LTS release	Long-term support, performance improvements
PostgreSQL	Open-source, reliable	Production-proven, supports large datasets
Elasticsearch	Fast, scalable	Industry-standard for search
Redis	Fast, in-memory	Distributed caching and messaging
S3	Scalable, cost-effective	Storage tiering support

2.13 Approval Sign-off

Role	Name	Date	Signature
Lead Engineer			Pending
DevOps Engineer			Pending
Security Lead			Pending

| Client Representative |
| | Pending |

Disclaimer: This document is for planning purposes only. All technical decisions should be validated through proof-of-concept development. Hardware requirements may vary based on specific workload and usage patterns. Consult with Nuxeo official documentation for the most up-to-date compatibility matrices.